INCH-POUND

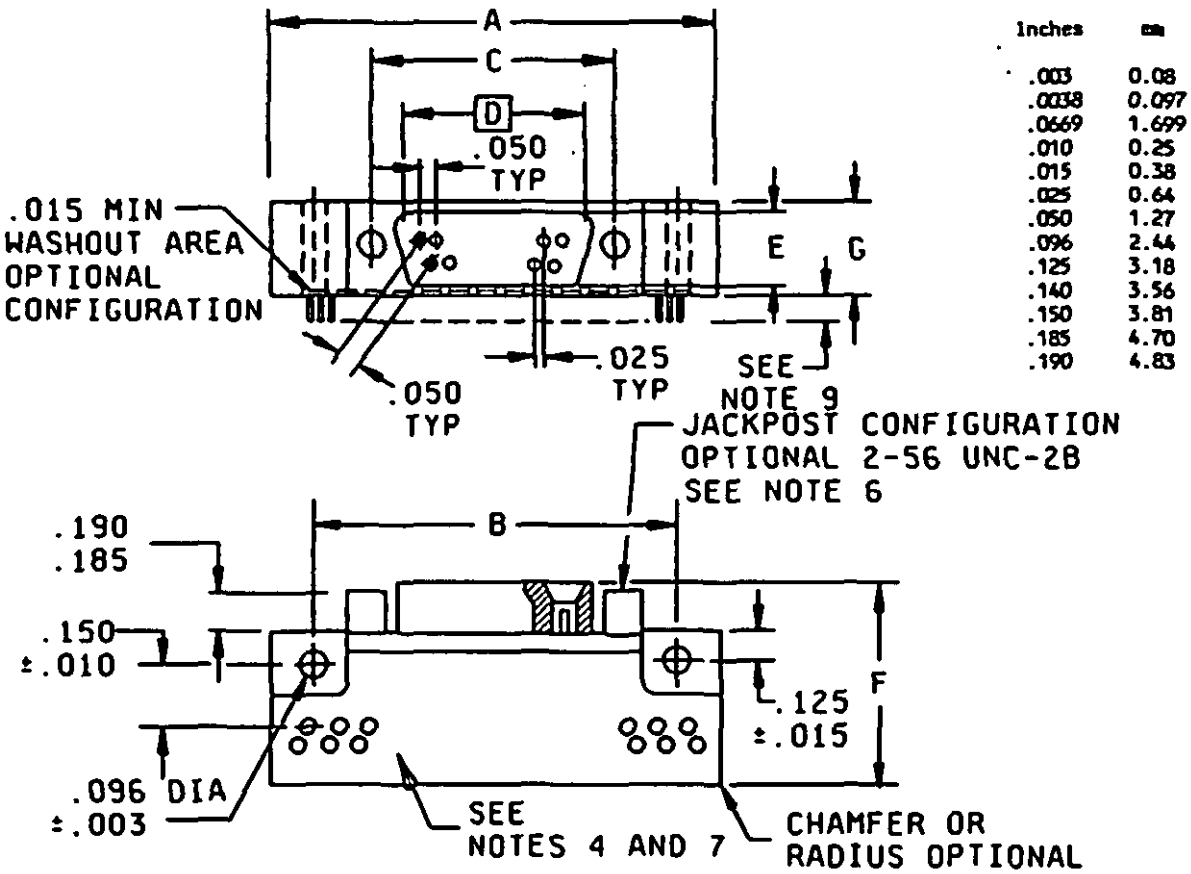
MIL-C-83513/16A
14 Oct 1993
SUPERSEDING
MIL-C-83513/16
27 February 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, PIN CONTACTS, 2 ROW, SOLDER TYPE, STANDARD PROFILE, 9 THROUGH 37 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83513.

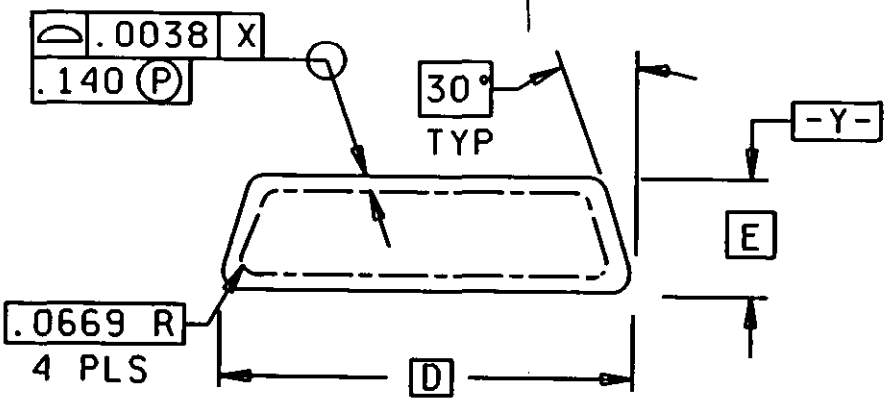


Inches	mm
.003	0.08
.0038	0.097
.0669	1.699
.010	0.25
.015	0.38
.025	0.64
.050	1.27
.096	2.44
.125	3.18
.140	3.56
.150	3.81
.185	4.70
.190	4.83

FIGURE 1. Connector, plug, .050 spacing.

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DETAIL OF INTERFACE

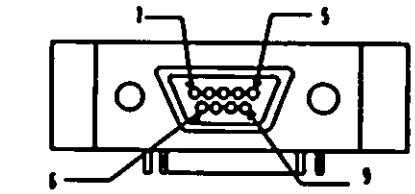
Number of contacts	A Max	B ±.007 (0.18 mm)	C ±.005 (0.13 mm)	D	E	F Max	G Max
9	1.390 (35.31)	1.150 (29.21)	.565 (14.35)	.3338 (8.48)	.1848 (4.70)	.465 (11.81)	.325 (8.26)
15	1.540 (39.12)	1.30 (33.02)	.715 (18.16)	.4838 (12.29)	.1848 (4.70)	.465 (11.81)	.325 (8.26)
21	1.690 (42.93)	1.450 (36.83)	.865 (21.97)	.6338 (16.10)	.1848 (4.70)	.465 (11.81)	.325 (8.26)
25	1.790 (45.47)	1.550 (39.37)	.965 (24.51)	.7338 (18.64)	.1848 (4.70)	.465 (11.81)	.325 (8.26)
31	2.040 (51.82)	1.800 (45.72)	1.115 (28.32)	.8838 (22.45)	.1848 (4.70)	.465 (11.81)	.325 (8.26)
37	2.340 (59.44)	2.100 (53.34)	1.265 (32.13)	1.0338 (26.26)	.1848 (4.70)	.465 (11.81)	.325 (8.26)

NOTES:

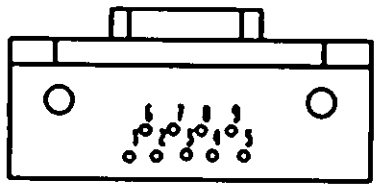
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513. Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
5. Metal shell shall be of material in accordance with MIL-C-83513 for class M parts.
6. Jackpost (permanently attached), when specified: Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.
7. Separately molded plastic body (if used) shall conform to the requirements of type GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.
8. Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.
9. Termination lengths available: .109 (2.77 mm), .140 (3.56 mm) or .172 (4.37 mm). The tolerance shall be ±.015 (0.381 mm) for all termination lengths.

FIGURE 1. Connector, plug, .050 spacing - Continued.

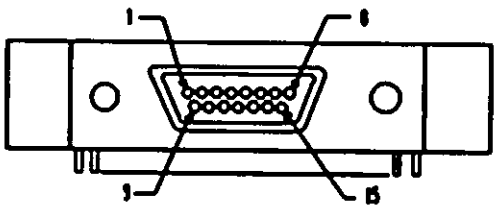
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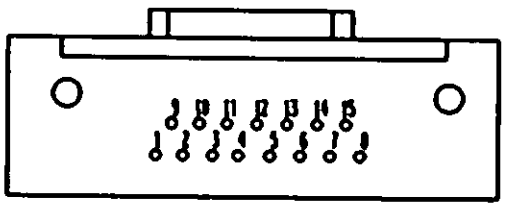


TERMINATION VIEW - RIGHT ANGLE
STANDARD PROFILE

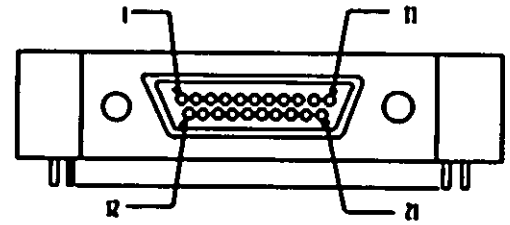


9 CAVITY

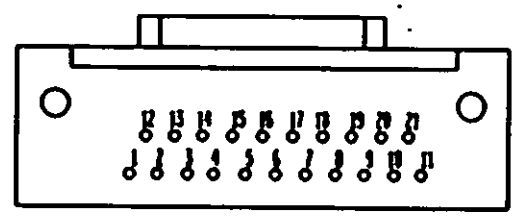




15 CAVITY



TERMINATION VIEW - RIGHT ANGLE
STANDARD PROFILE

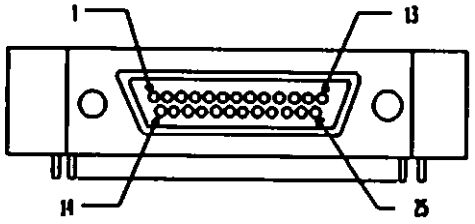


21 CAVITY

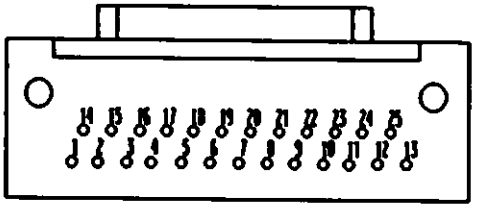
NOTE:
Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement.

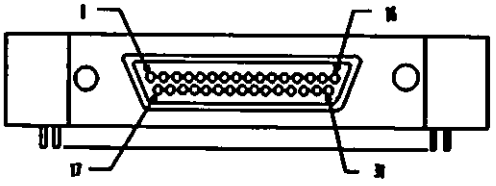
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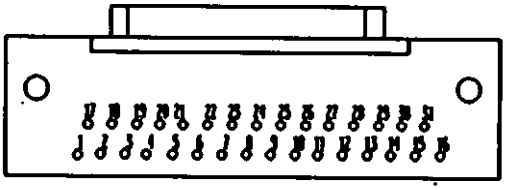
TERMINATION VIEW - RIGHT ANGLE
STANDARD PROFILE



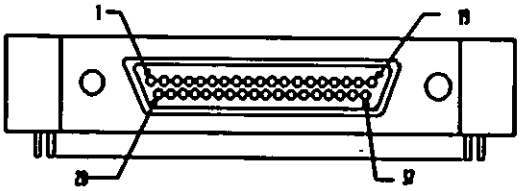
25 CAVITY



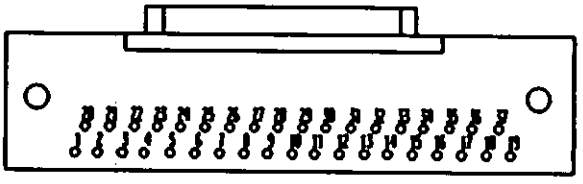
TERMINATION VIEW - RIGHT ANGLE
STANDARD PROFILE



31 CAVITY



TERMINATION VIEW - RIGHT ANGLE
STANDARD PROFILE

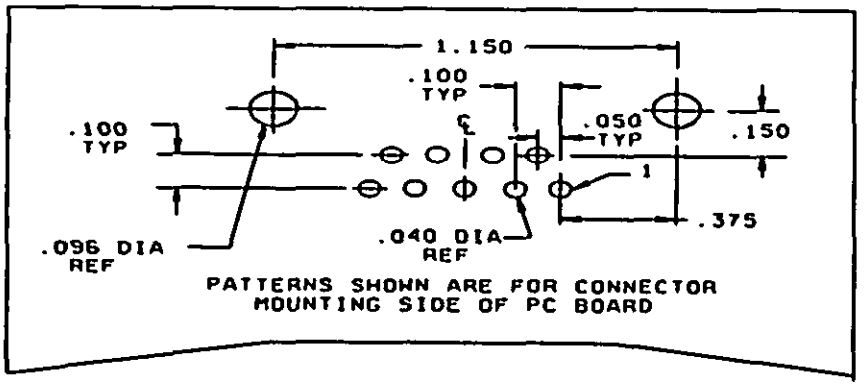


37 CAVITY

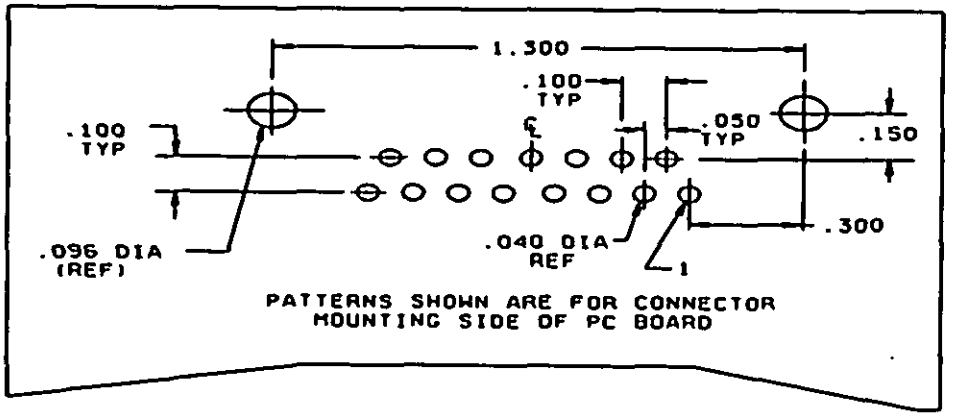
NOTE:
Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part.

FIGURE 2. Insert arrangement - Continued.

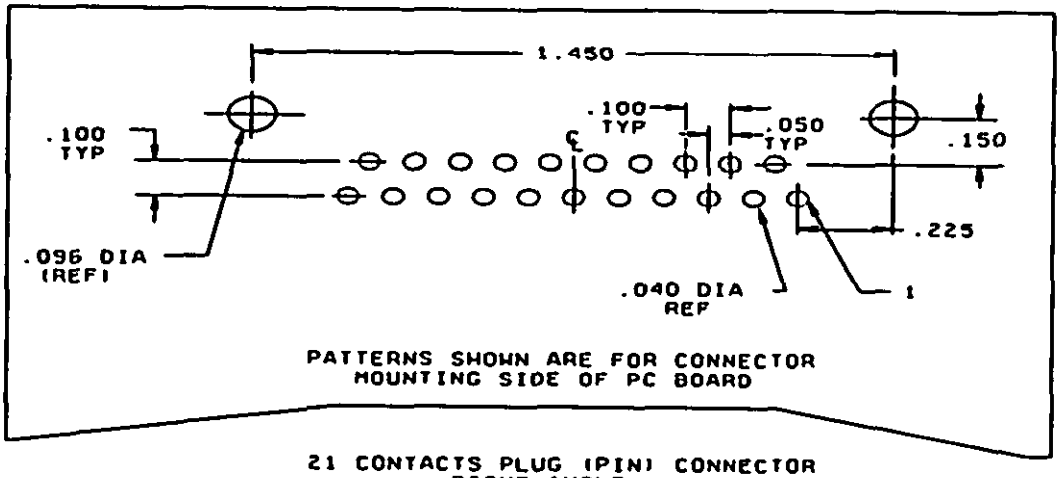
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9 CONTACTS PLUG (PIN) CONNECTOR
-RIGHT ANGLE-



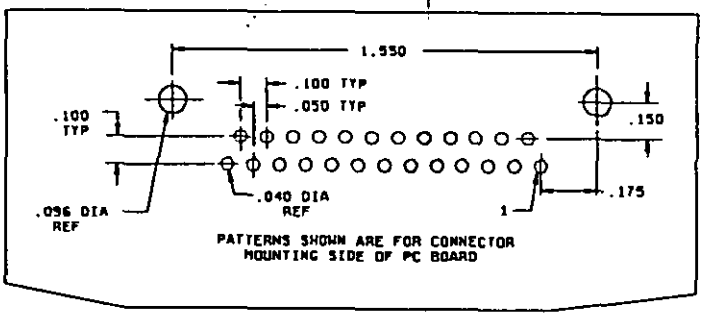
15 CONTACTS PLUG (PIN) CONNECTOR
-RIGHT ANGLE-



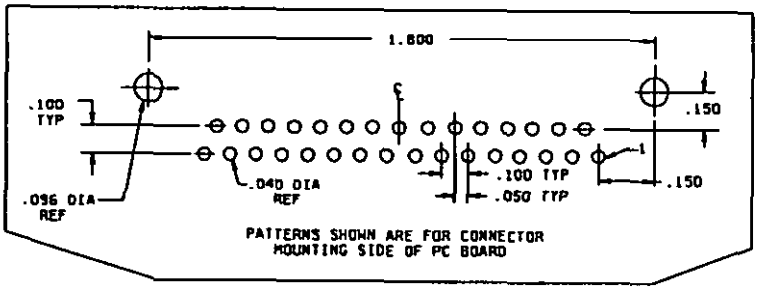
21 CONTACTS PLUG (PIN) CONNECTOR
-RIGHT ANGLE-

FIGURE 3. Layout arrangement.

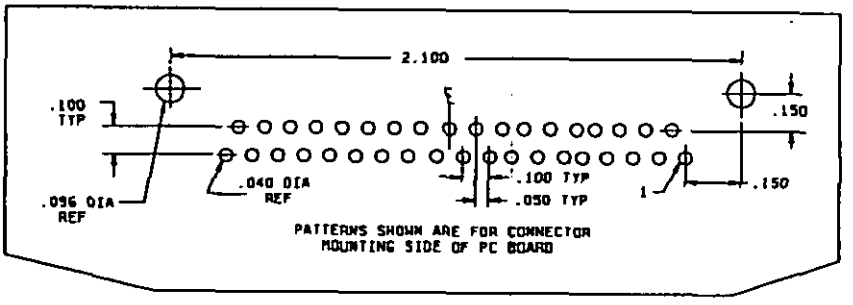
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25 CONTACTS PLUG (PIN) CONNECTOR
-RIGHT ANGLE-



31 CONTACTS PLUG (PIN) CONNECTOR
-RIGHT ANGLE-



37 CONTACTS PLUG (PIN) CONNECTOR
-RIGHT ANGLE-

Inches	mm	Inches	mm	Inches	mm
.040	1.02	.175	4.44	1.300	33.02
.050	1.27	.225	5.72	1.450	36.83
.096	2.44	.300	7.62	1.550	39.37
.100	2.54	.375	9.53	1.800	45.72
.150	3.81	1.150	29.21	2.100	53.34

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Pattern shown is for connector mounting side of PC board.

FIGURE 3. Layout arrangement - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2.

Current rating, maximum: 3 amperes per contact.

Materials:

Termination organization area: Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513.

Shell: The requirements for shell materials shall be in accordance with MIL-C-83513.

Plastic body or plastic molding: Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519.

Jackpost: Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35.

Wire termination pins: Wire termination pins shall conform to QQ-W-343, no. 24 AWG copper.

Mating connector: Shall conform to MIL-C-83513/2 and MIL-C-83513/4.

Plating of termination leads: Solder dipping of termination leads will be accomplished in SN60 PB40 or SN63 PB37 in accordance with QQ-S-571.

Part or Identifying Number (PIN): Consists of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code for the shell finish and the hardware.

M83513/16-	D	01	C	P
Specification sheet number	Insert arrangements (see figure 2) A = 09 B = 15 C = 21 D = 25 E = 31 F = 37	Termination length 01 = .109 02 = .140 03 = .172	Shell finish C = Cadmium N = Electroless nickel (space application)	Hardware N = No jackpost P = Jackpost attached

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - MI
Navy - AS, SH
Air Force - 17, 99

User activities:
Army - AT, ME
Navy - CG, MC

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project: 5935-3921-07)

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